

熵

定义及性质

- 若 X 是取值于 S 上的随机变量，定义其熵为

$$H(X) = - \sum_{s \in S} p_s \log p_s$$

- 若 X, Y 是两个随机变量，定义其熵为

$$H(X, Y) = - \sum_{x, y} \mathbb{P}(X = x, Y = y) \log \mathbb{P}(X = x, Y = y)$$

- 定义事件 E 下 X 的条件熵为

$$H(X|E) = - \sum_x \mathbb{P}(X = x|E) \log \mathbb{P}(X = x|E)$$

- 定义 X 对随机变量 Y 的条件熵为

$$H(X|Y) = \mathbb{E}_Y H(X|Y) = - \sum_{x, y} \mathbb{P}(Y = y) \mathbb{P}(X = x|Y = y) \log \mathbb{P}(X = x|Y = y)$$

- 若 $|S| = n$ ，由 Jensen 不等式易得

$$H(X) \leq \log n$$

- $H(X) \geq 0$ ，取 "=" iff $\exists p_s = 1$
- $H(X, Y) = H(X) + H(Y|X)$
- $H(X|Y) \leq H(X)$ ，取 "=" iff X, Y 独立
- (次可加性) $H(X, Y) \leq H(X) + H(Y)$ ，取 "=" iff X, Y 独立

熵的应用

Thm Let $\mathcal{F}$ be a collection of subsets of $[n]$ and p_i be the fraction of subsets in $\mathcal{F}$ that contains i . Then

$$\log |\mathcal{F}| \leq \sum_i H(p_i)$$

where $H(p_i) = -p_i \log p_i - (1 - p_i) \log(1 - p_i)$

Sol

Let S be a uniformly random sampling on $\mathcal{F}$.

Define $X_i = 1_{\{i \in S\}}$, and $X = (X_1, \dots, X_n)$.

S and X contain the same information, or more exactly, S and X uniquely determine each other mutually.

Then

$$\log |\mathcal{F}| = H(X) \leq \sum_i H(X_i) = \sum_i H(p_i)$$

Thm Let $S_1, \dots, S_k$ be subsets of $[n]$, and suppose for every pair of distinct $A, B \subset [n]$, $\exists i$ s. t. $|S_i \cap A| \neq |S_i \cap B|$. Determine a lower bound of k using entropy.

Sol

Let X be a uniformly sampled random subset of $[n]$.

$(|X \cap S_i|)_{1 \leq i \leq k}$ and X uniquely determine each other mutually.

Then

$$H(X) = H((|X \cap S_i|)_{1 \leq i \leq k}) \leq \sum_{i=1}^k H(|X \cap S_i|)$$

$|X \cap S_i|$ follows a binomial distribution. The problem is then reduced to estimate the upper bound of $H(Y)$, where $Y \sim B(n, \frac{1}{2})$

By CLT, we can get an estimation

$$H(Y) = \frac{1}{2} \log \left(\frac{\pi e}{2} n \right) + O \left(\frac{1}{n} \right) = \frac{1}{2} \log n + O(1)$$

Back to inequality, we have

$$\begin{aligned} n = H(X) &\leq k \cdot \left(\frac{1}{2} \log n + O(1) \right) \\ &\Rightarrow \\ k &\geq \frac{2n}{\log n + O(1)} \end{aligned}$$

Similar to determinant of a square matrix, we define the permanent of square matrix A as

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{a_i \in \sigma} a_i$$

Especially for 0 – 1 matrix A , $\text{per}(A) = \#\{\text{perfect matchings in corresponding bipartite graph}\}$.

Thm Define $d_i = \sum_{j=1}^n a_{ij}$ where $A = (a_{ij})$ is a 0 – 1 matrix. Then

$$\text{per}(A) \leq \prod_{i=1}^n (d_i!)^{1/d_i}$$

proof

Let σ be a uniform permutation of $[n]$ conditioned on $a_{i\sigma(i)} = 1, \forall i$.

Then $H(\sigma) = \log \text{per}(A)$

Let τ be a uniform permutation of S_n

By additivity of entropy, we get

$$\begin{aligned}
H(\sigma) &= \sum_{i=1}^n H(\sigma(\tau_i) | \sigma(\tau_1), \dots, \sigma(\tau_{i-1})) \\
&= \sum_{i=1}^n \mathbb{E}_{\tau} H(\sigma(\tau_i) | \sigma(\tau_1), \dots, \sigma(\tau_{i-1}))
\end{aligned}$$

where $H(\sigma(i) | \sigma(1), \dots, \sigma(i-1))$ stands for the entropy of determining the i^{th} row with former $i-1$ rows determined.

For row i , when it is connected at step j , $j-1$'s different columns were used in former $j-1$ steps. Thus, number of available steps for row i is at least $d_i - (j-1)$.

i. e.

$$H(\sigma(\tau_j) | \sigma(\tau_1), \dots, \sigma(\tau_{j-1})) \leq \log(d_i - j + 1)$$

where $\tau_j = i$.

Taking expectation for τ , we get

$$\begin{aligned}
\mathbb{E}_{\tau} H(\sigma(i) | \dots) &= \mathbb{E}_{\tau} H(\sigma(\tau_j) | \sigma(\tau_1), \dots, \sigma(\tau_{j-1})) \\
&\leq \frac{1}{d_i} (\log 1 + \dots + \log d_i) = \frac{1}{d_i} \log d_i!
\end{aligned}$$

The conclusion follows then.